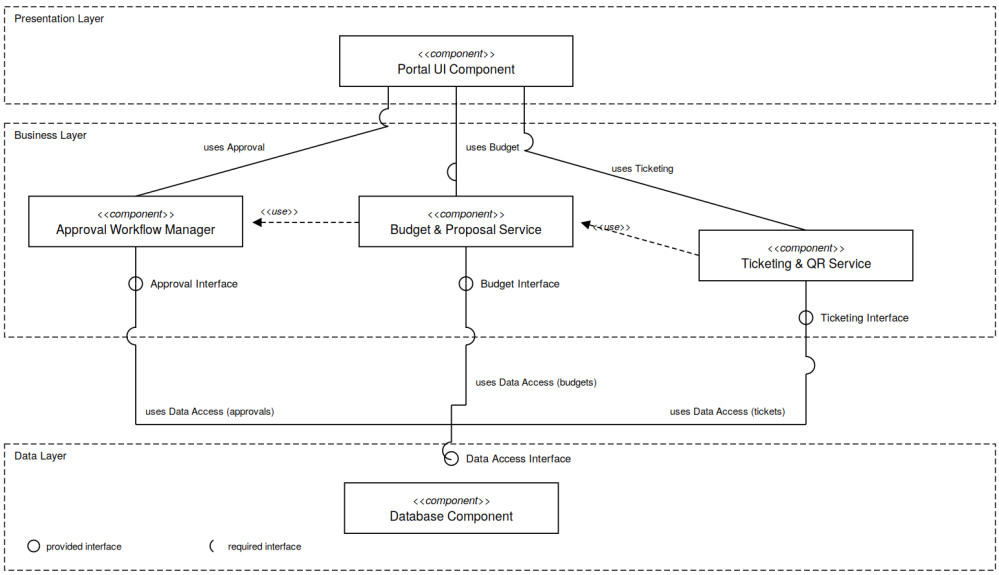
Lab 3: Component Modelling & Architectural Pattern Selection

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Problem Statement #10: Student Club Event Ticketing & Budget Portal

Deliverable 1: Component Diagram



Deliverable 2: Written Justification

Architecture Selection: I chose Layered Architecture for the Student Club Event Ticketing & Budget Portal System.

Reason 1: Multi-stage approval maps naturally onto layers

The portal's core requirement (FR-001) is a strict, sequential approval chain from Faculty Coordinator to Finance Office to Dean. A layered structure keeps this workflow logic isolated in a single business layer, separate from the presentation layer (portal UI) and the data layer (proposal and budget records). This means the approval sequence can be changed or extended without touching how the club lead fills out the form or how records are stored, which keeps the system easy to maintain as approval rules evolve.

Reason 2: Small team, predictable load, no need for independent scaling

This is a single-institution academic tool used by club leads and a handful of admin roles, not a system with unpredictable or highly variable traffic across different functions. Microservices would add operational overhead (separate deployments, network calls, service discovery) that this scenario does not need. A layered monolith is simpler to build, test, and deploy for a system of this scale, while still keeping concerns cleanly separated.

Security Advantage

Because all financial and role-based access logic (NFR-002: RBAC, AES-256 encryption at rest, TLS 1.3 in transit) sits in the business layer, the presentation layer never directly touches the database. Every request from the portal UI must pass through the Approval Workflow Manager or Budget & Proposal Service, which enforce role checks before any query reaches the Database Component. This gives a single, consistent place to apply and audit access control rather than scattering permission checks across the UI.

Performance Benefit

The QR ticket scanner's 100ms validation requirement (NFR-001) is met by keeping the Ticketing & QR Service as a thin, focused component that only needs to check a signed QR payload against the database, without routing through the approval or budget logic. Because layers only communicate through well-defined interfaces, the scanning path stays short and predictable even under peak check-in load.